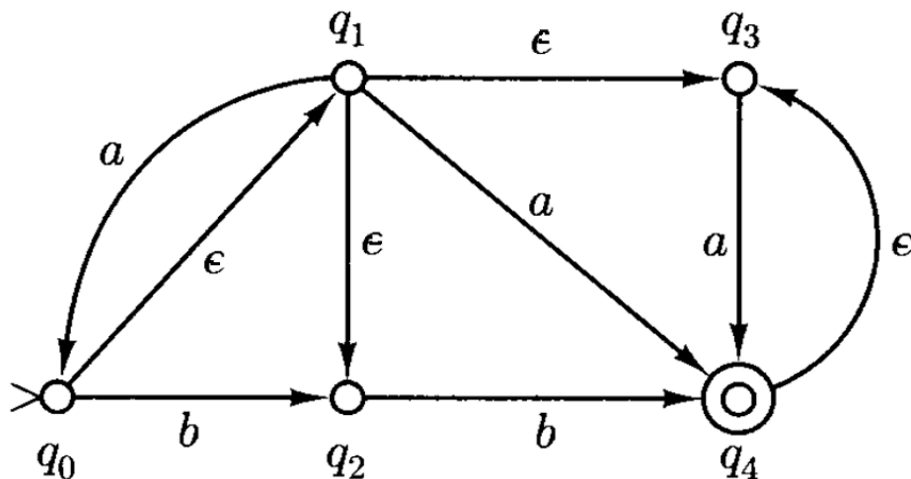


CSE331 Assignment 1 Part A

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- Deadline for entire Assignment 1 Part A submission:
8 July, 2026, Wednesday, 11:59 pm.
 - The submitted document should have clear and legible scans.
1. Draw direct (No cross product rule) DFA accepting the following languages, $\Sigma = \{0,1\}$:
 - a. Has both even number of 0's and even number of 1's
 - b. Binary representation of the string is divisible by 3 or 5
 - c. Non-empty string such that there can be at most two consecutive 0's, and the pair should have at least one 1 in both its front and end
 - d. Contains substring 101 at most two times
 - e. Each 1 is followed by at least one 0 but cannot end with 100
 - f. When substring 00 is found, substring 11 should exist as well and vice versa.
 - g. The absolute value of the difference between the number of 0's and the number of 1's is odd.
 2. Read the following description and draw DFA, $\Sigma = \{0,1\}$, Show at least 8 states along with transitions:
 $L_1 = \{w \mid w = 1^n, n \text{ is a multiple of } 3\}$
 $L_2 = \{w \mid w = 0^m, m \text{ is 2 more than multiple of } 4\}$
 $L_3 = \{w \mid w \text{ is a non-empty string containing 0 in every position which is a multiple of } 3\}$ [Here, position starts from 1]
 $L_4 = \{w \mid w \text{ contains substring } 1011\}$
 - A. $L_2 \cup \bar{L}_3$
 - B. $L_3 \cap L_4$
 - C. $L_1 \circ L_2$
 - D. $L_3 \oplus L_4$ [Optional]
 3. Convert the following NFA into DFA, you need to show detailed steps. Also answer the following questions:



- a. How many maximum states can be found when converting the given NFA to DFA using subset construction method?
- b. What is the maximum number of accepting states and rejecting states for the converted NFA?
- c. What is the ϵ -closure of the state q_0 ?
- d. **$\delta(\{q\}, a)$ outputs the set of states in which the NFA transitions when it is in state q and reads the symbol a .** According to this definition, what is $\delta(\{q_0, q_3\}, a)$ in the given NFA?